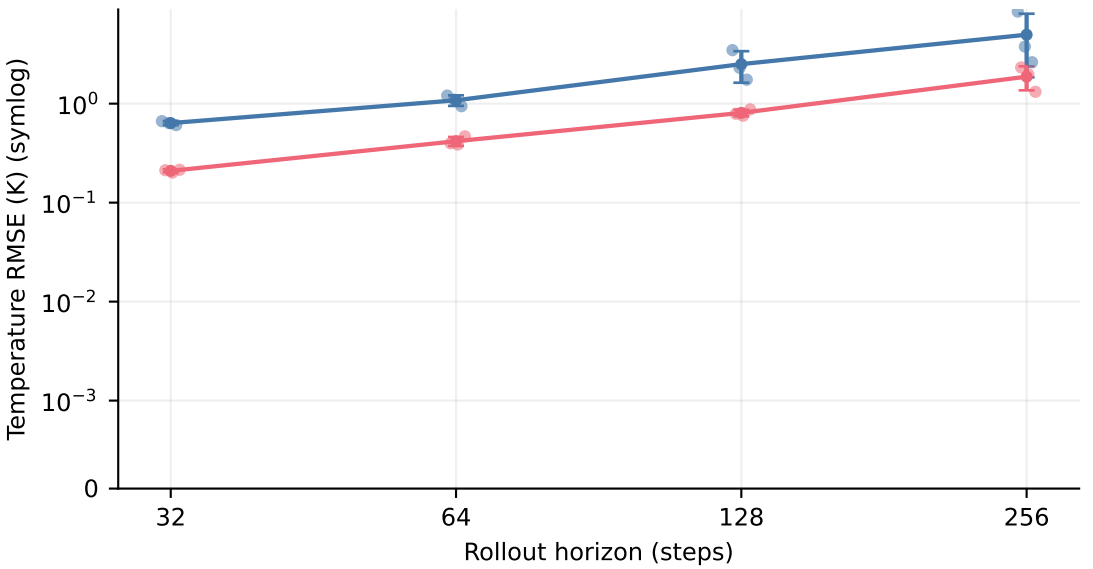
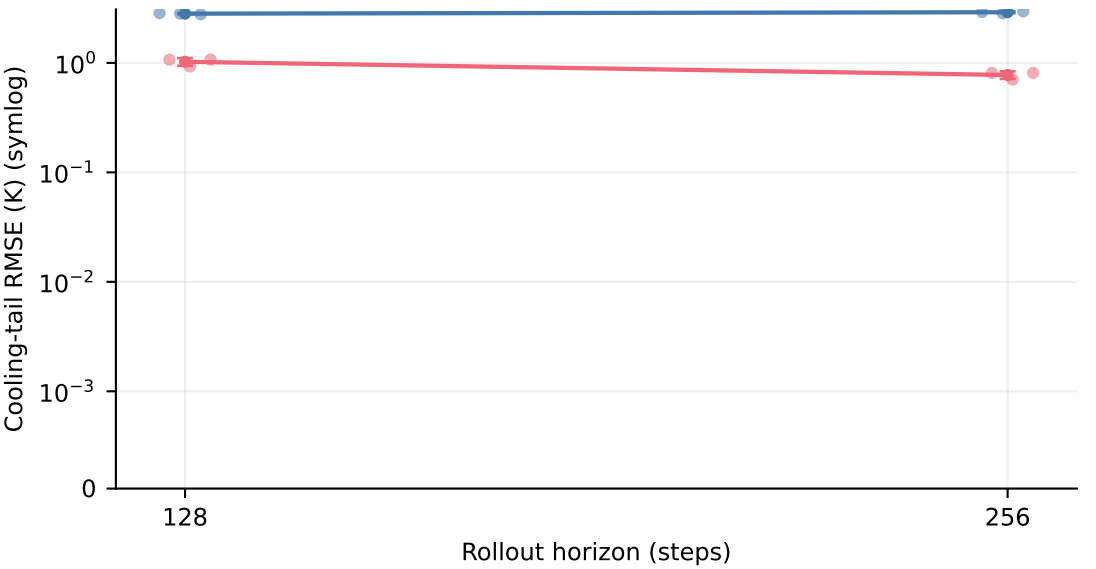
B: explicit-state update ablations

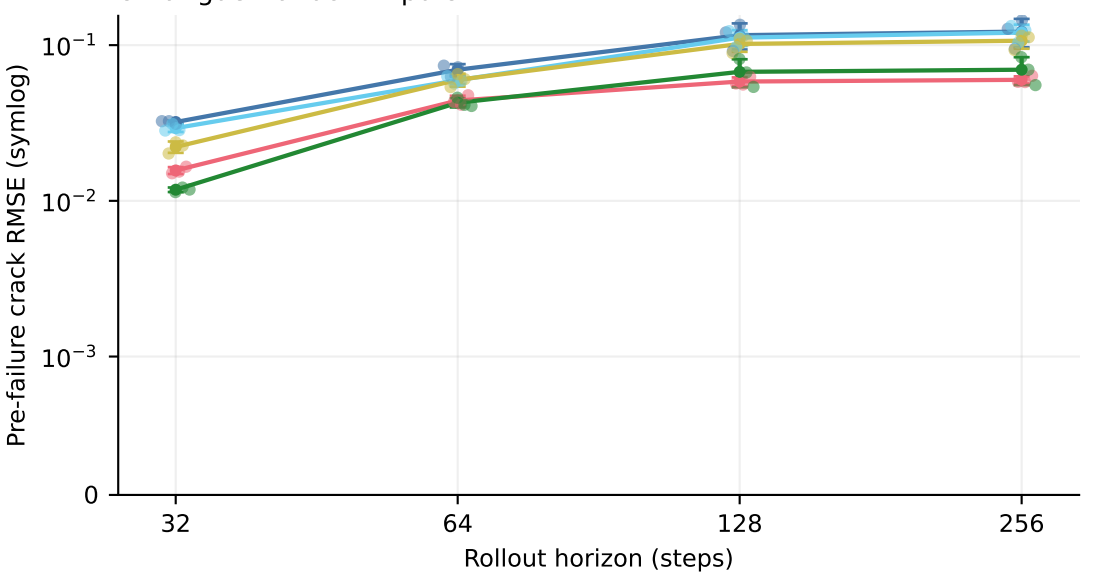
a Thermal: random inputs



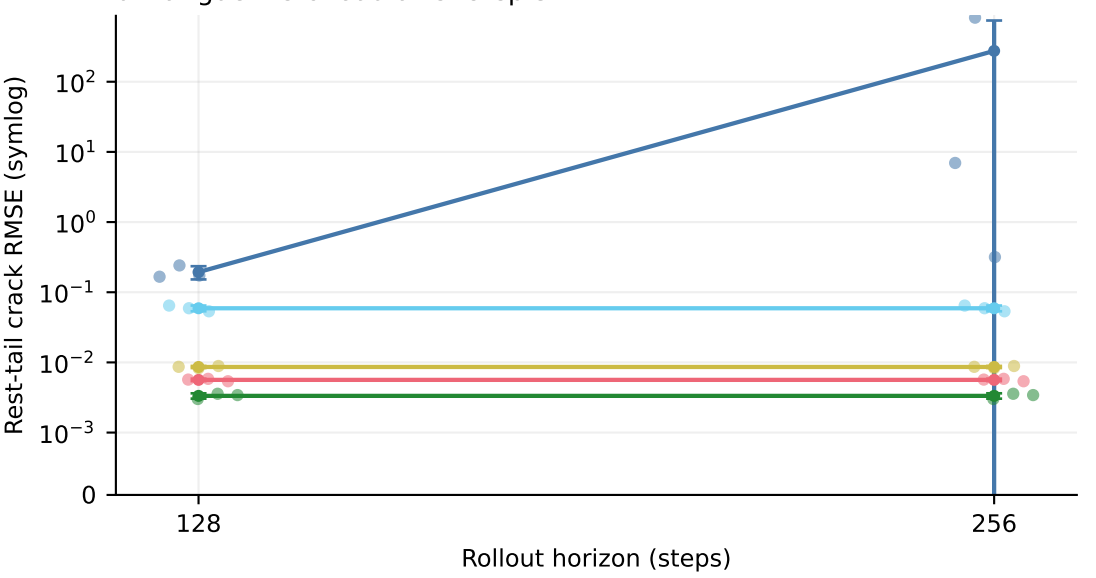
b Thermal: zero power after step 32



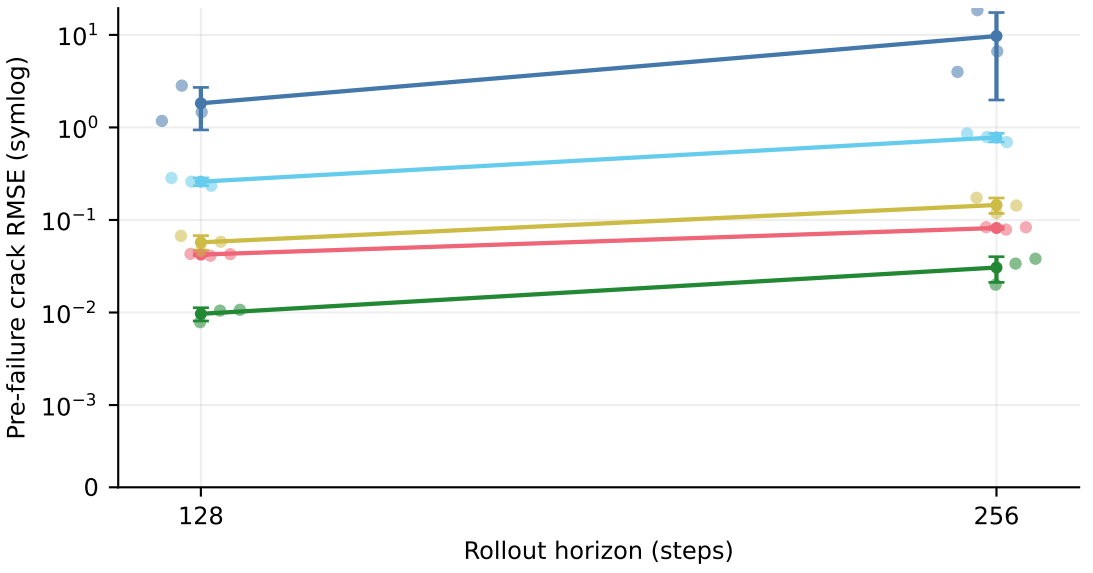
c Fatigue: random inputs



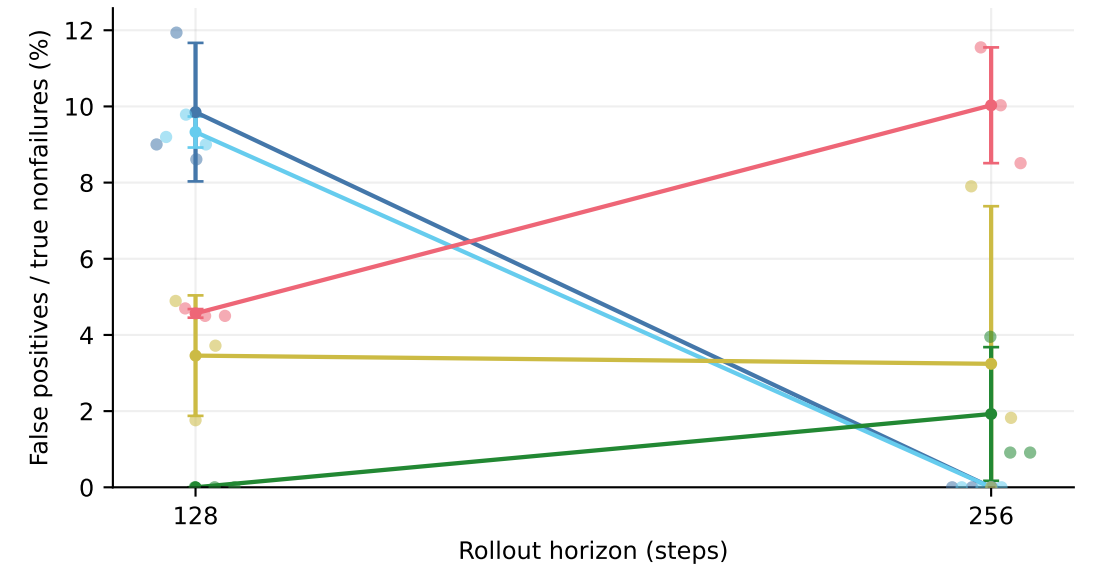
d Fatigue: zero load after step 32



e Fatigue: sustained low load



f Fatigue: low-load false positives



Free update
+ stress gate
+ positive increment
Full physical update / + cap
Full + cubic gate

Points: 3 seeds; lines and bars: mean +/- sample SD. No mean is plotted for incomplete seed sets.
Train horizon = 32; tests = 512 fixed episodes. SD bars below zero are clipped to the nonnegative domain; statistics unchanged.